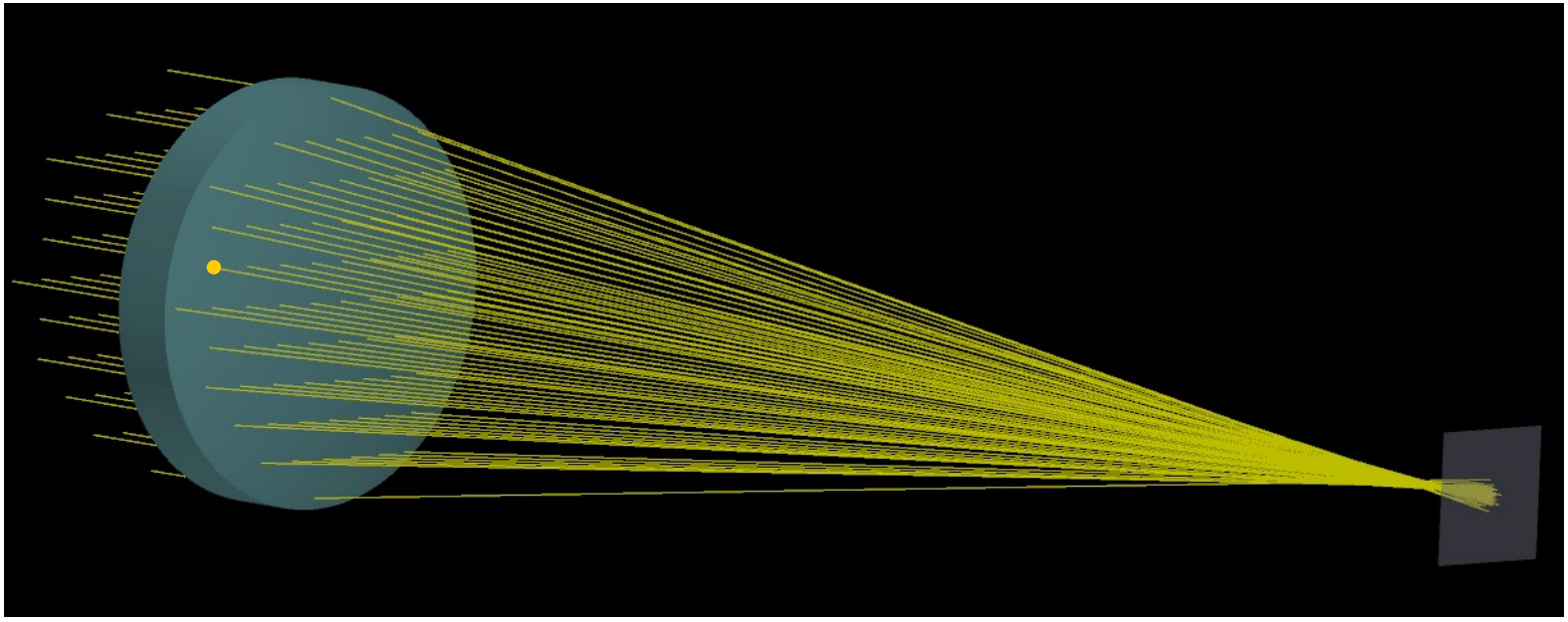
▪ Advantages of geometrical optics ▪

Vector diffraction: Solving Maxwell's equations (finite element method)

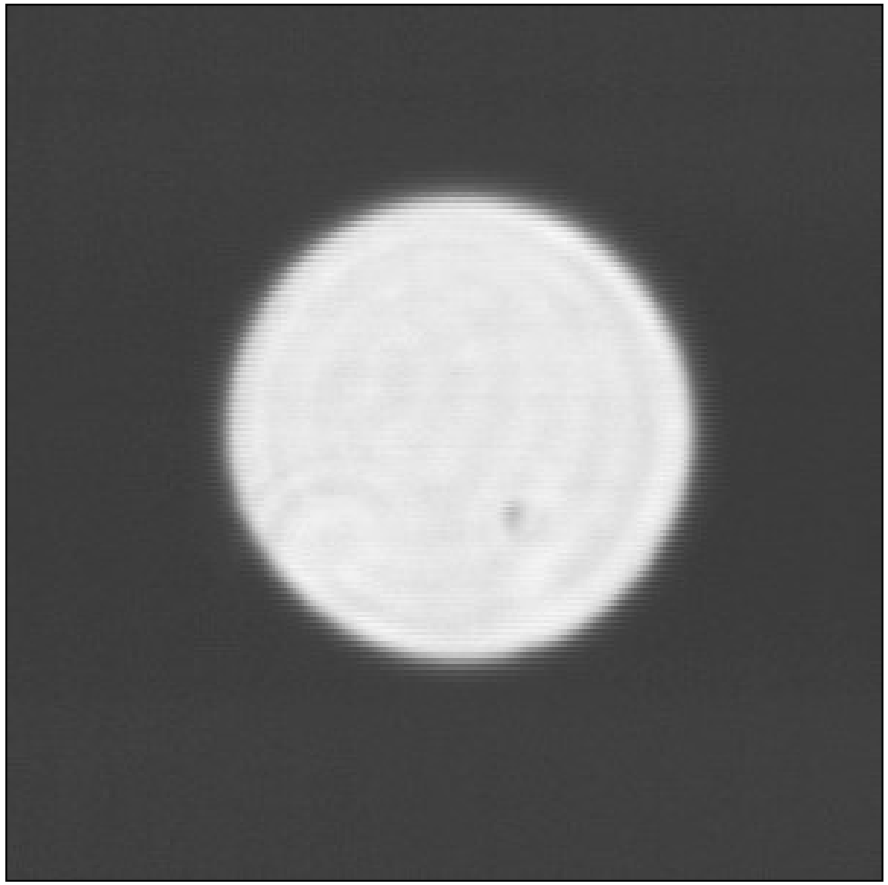
Scalar diffraction: Fresnel-Kirchhoff diffraction (rarely used ~10^8 sample points)

⇩

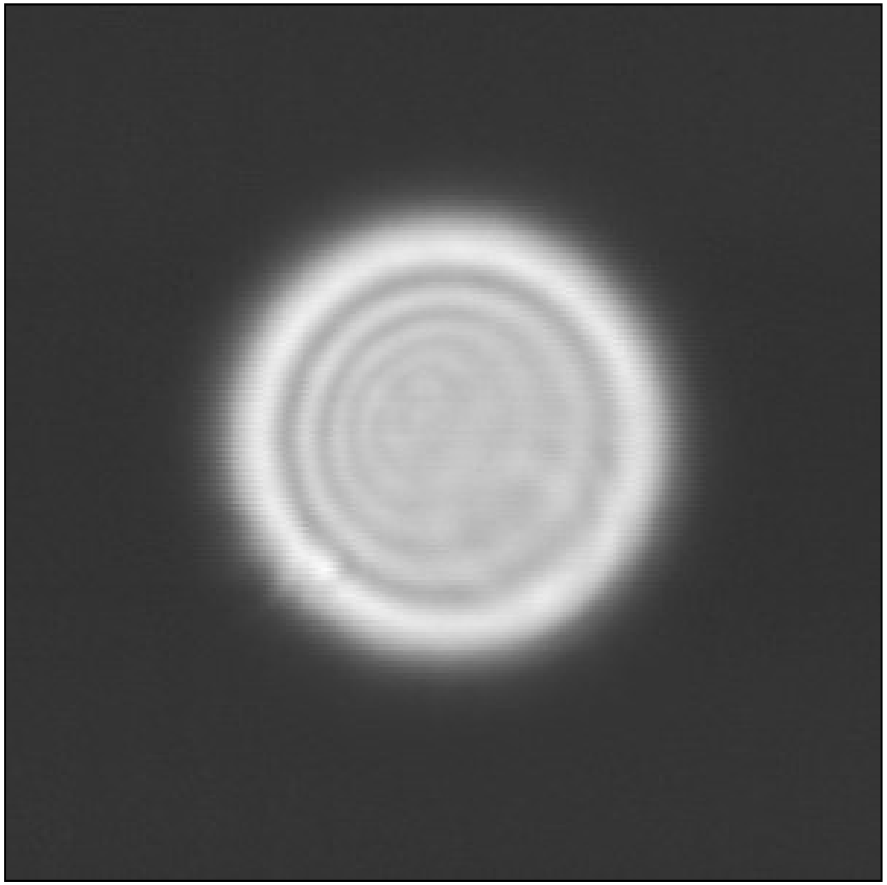
Fresnel approximation:	~10^6 sample points
Fraunhofer approximation:	~10^4 sample points
Geometrical optics:	~10^2 sample points
	(~100,000,000 intersection/sec)



▪ The limits of geometrical optics ▪

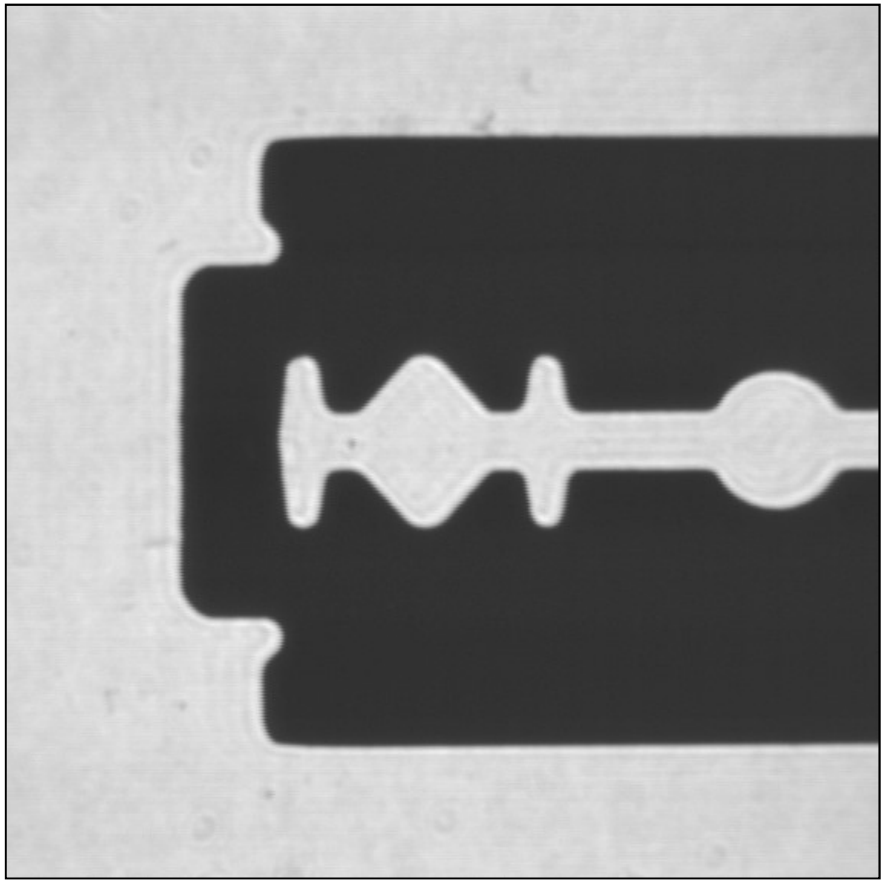


Geometrical optical shadow

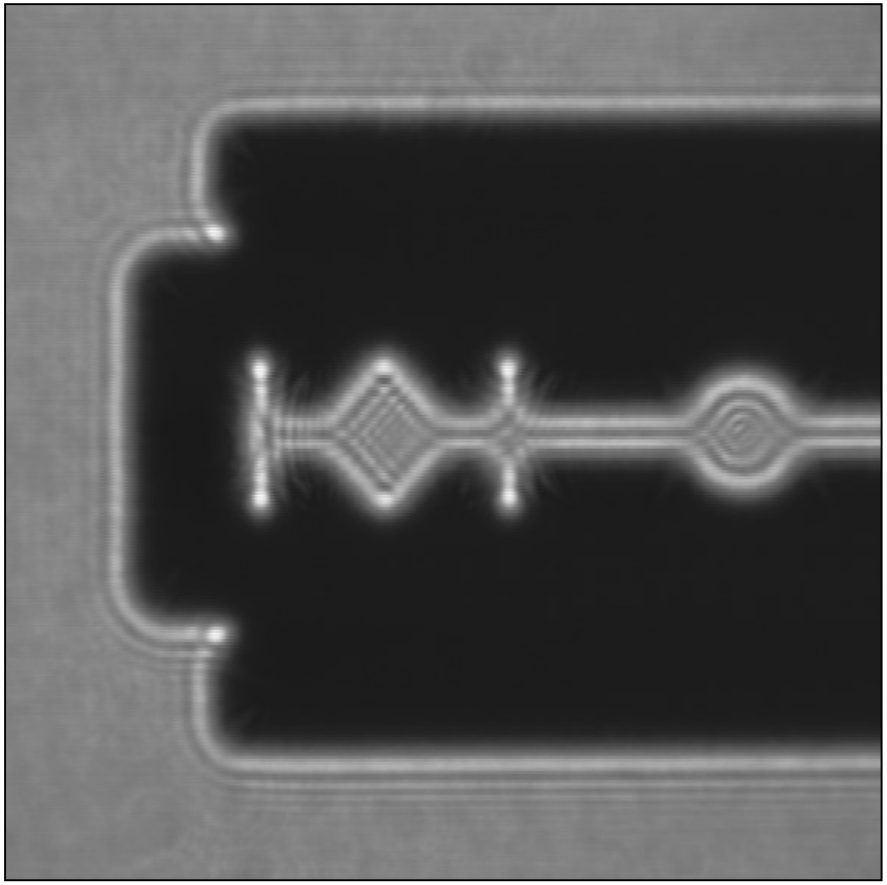


Fresnel diffraction

▪ The limits of geometrical optics ▪

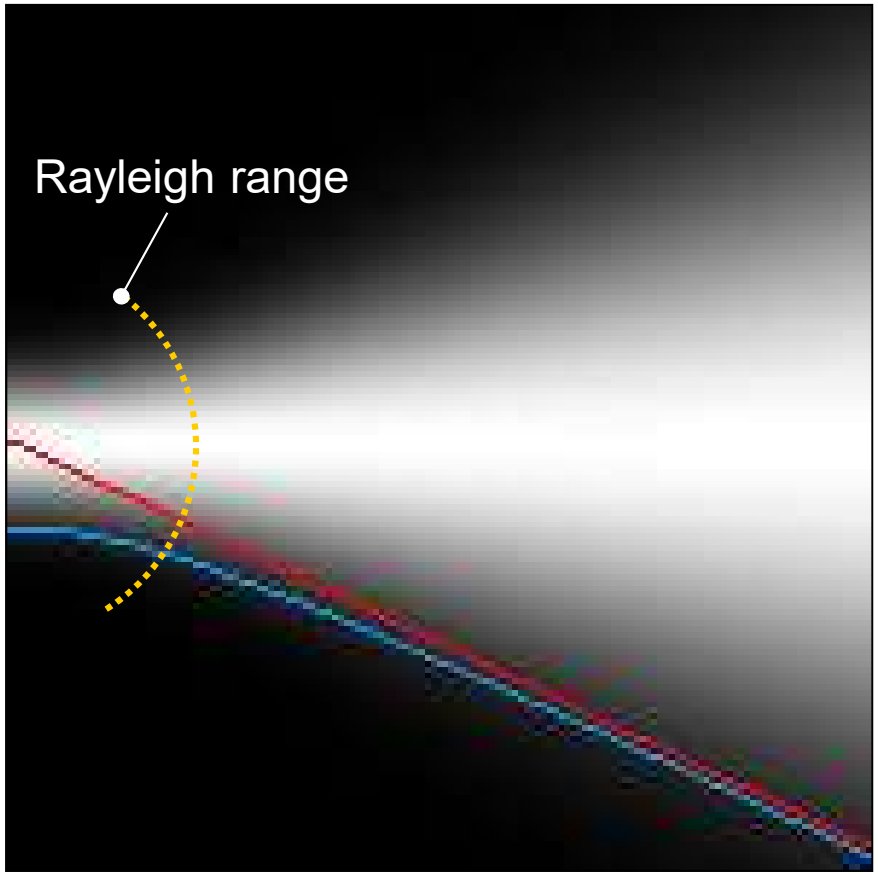


Geometrical optical shadow

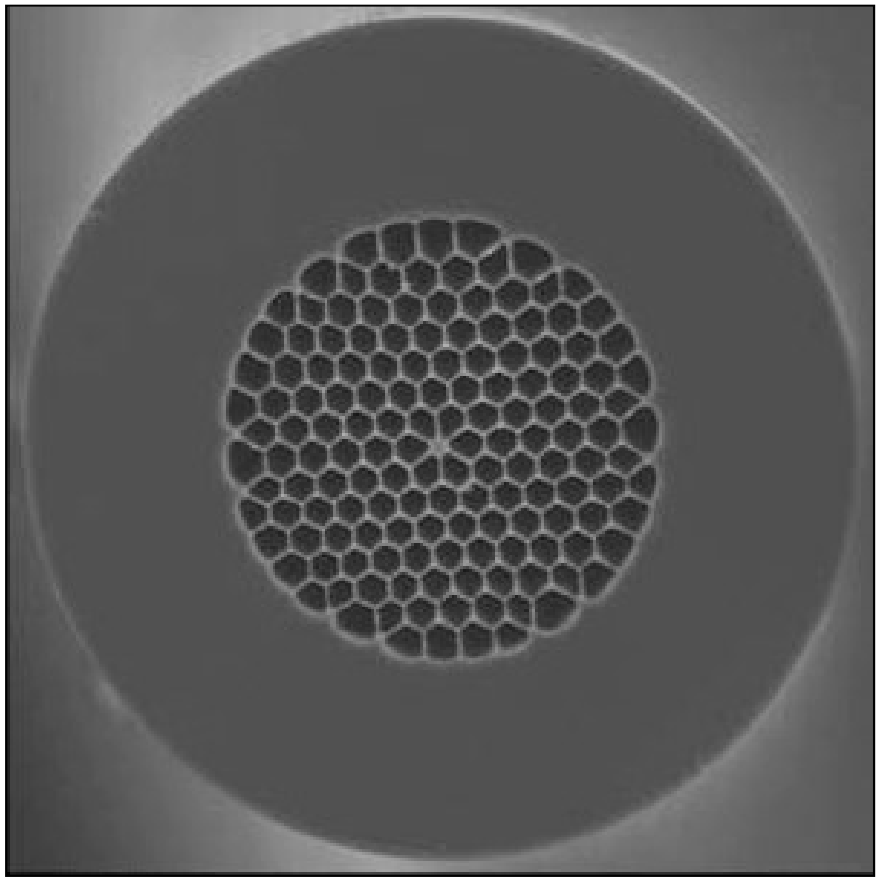


Fresnel diffraction

▪ The limits of geometrical optics ▪



Propagation of Gaussian beams



Photonic crystal

▪ The limits of geometrical optics ▪

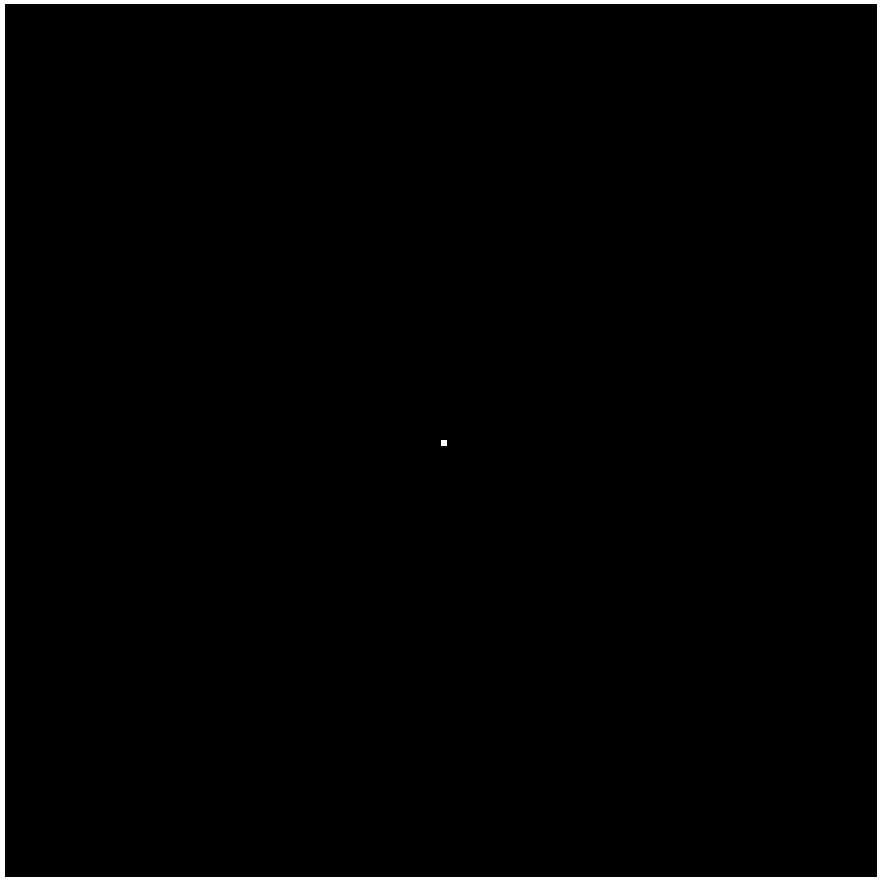


Image by geometrical optics

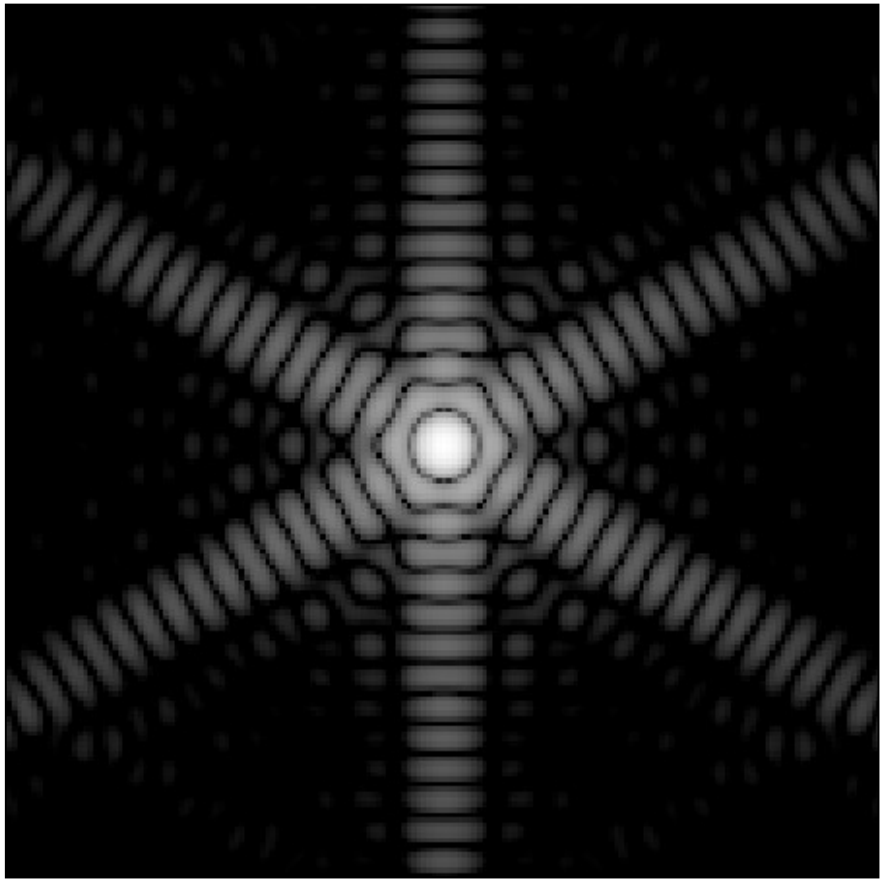
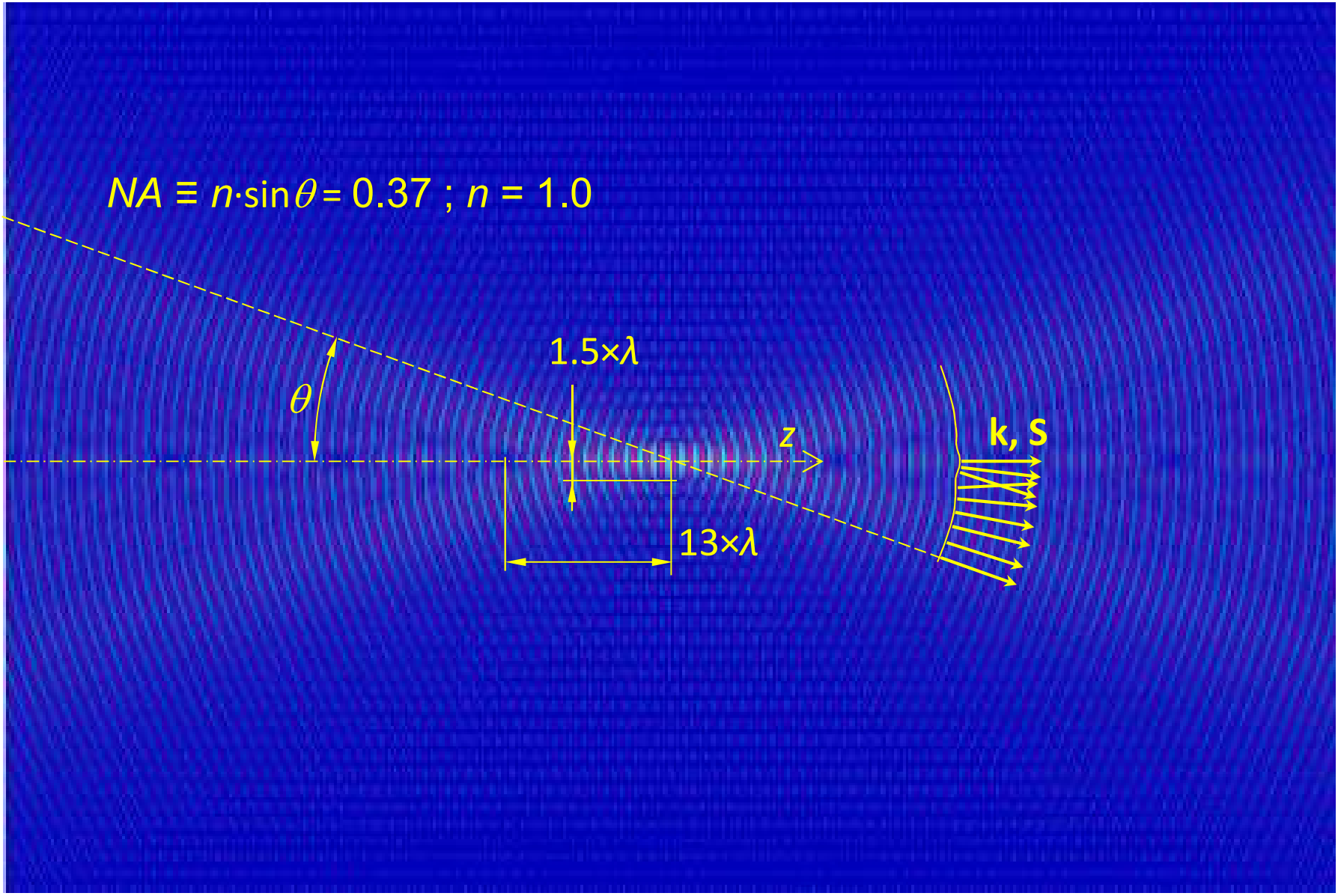
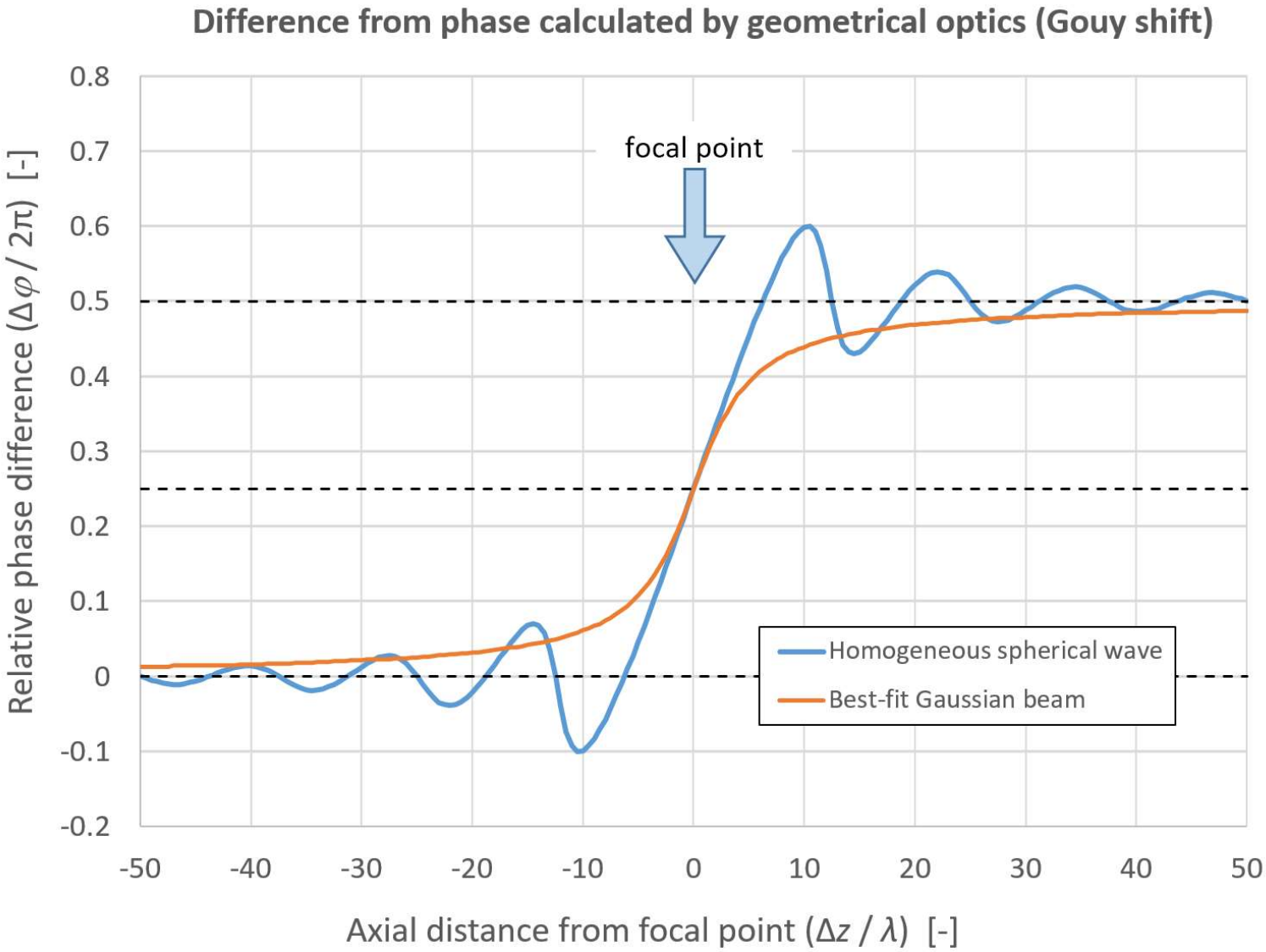


Image by Fraunhofer diffraction

▪ Wavefronts near focus ▪



▪ Phase changes near focus ▪



$$\frac{\lambda_{\mathrm{loc}} - \lambda}{\lambda} \approx \frac{\Delta\varphi/2\pi}{\Delta z/\lambda}$$

$$\left.\frac{\Delta\lambda_{\mathrm{loc}}}{\lambda}\right|_{\mathrm{max}} \approx +3.5\%$$

▪ Local plane wave approximation ▪

Slowly varying amplitude approximation

$$\frac{\left|\text{grad}(E_{0x})\right|}{E_{0x}} << k_0 \cdot \left|\text{grad}(S)\right| \quad \Rightarrow$$

Local wave vector and wavelength

$$\mathbf{k}_{lok} = k_0 \cdot grad(S)$$

$$\Rightarrow \quad \frac{\left|\text{grad}(E_{0x})\right| \cdot \lambda}{E_{0x}} << 2\pi$$

$$\Downarrow \qquad \Nearrow$$

Phase changes be not affected by E_{0x}

$$\left|\nabla^2 E_{0x}\right| << k_{lok}^2 \cdot E_{0x} \quad \Rightarrow \quad \left|\nabla S\right| = n \quad \Rightarrow$$

$$k_{lok} = k_0 \cdot n$$

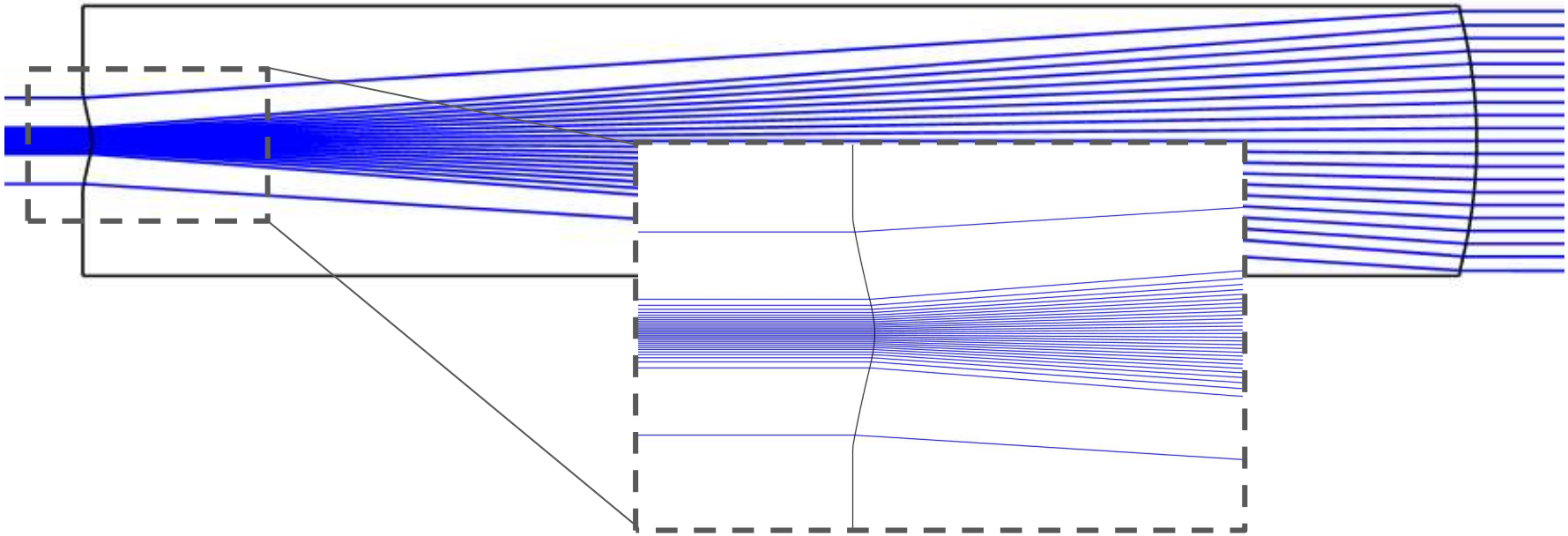
$$\lambda_{lok} = \lambda = \frac{\lambda_0}{n} \quad \Rightarrow \quad \frac{\left|\nabla^2 E_{0x}\right| \cdot \lambda^2}{E_{0x}} << 4\pi^2$$

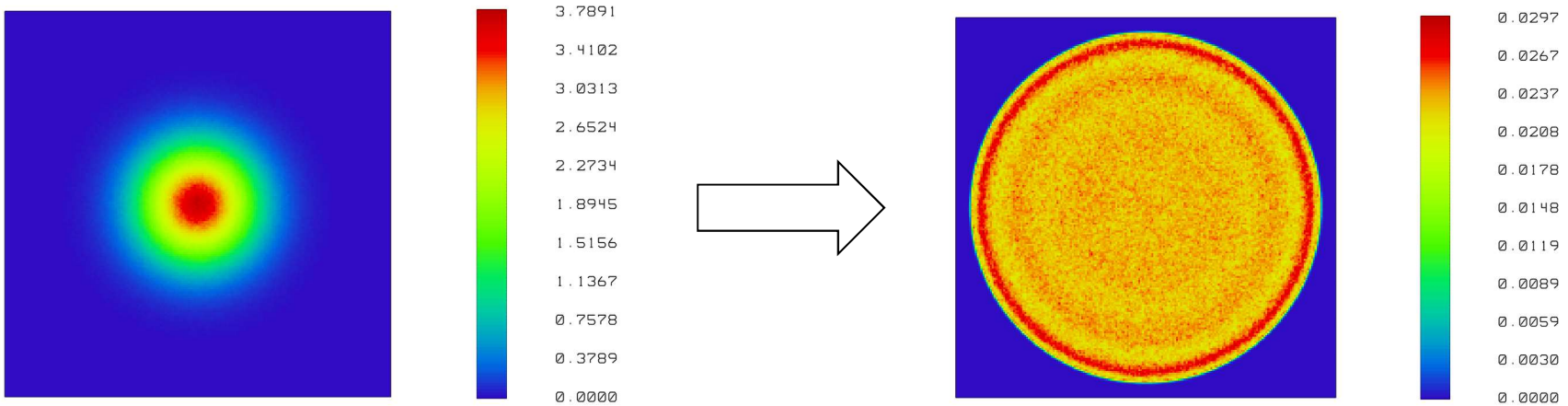
Approximation of short wavelengths: geometrical optics

▪ Wave equation in terms of the eikonal ▪

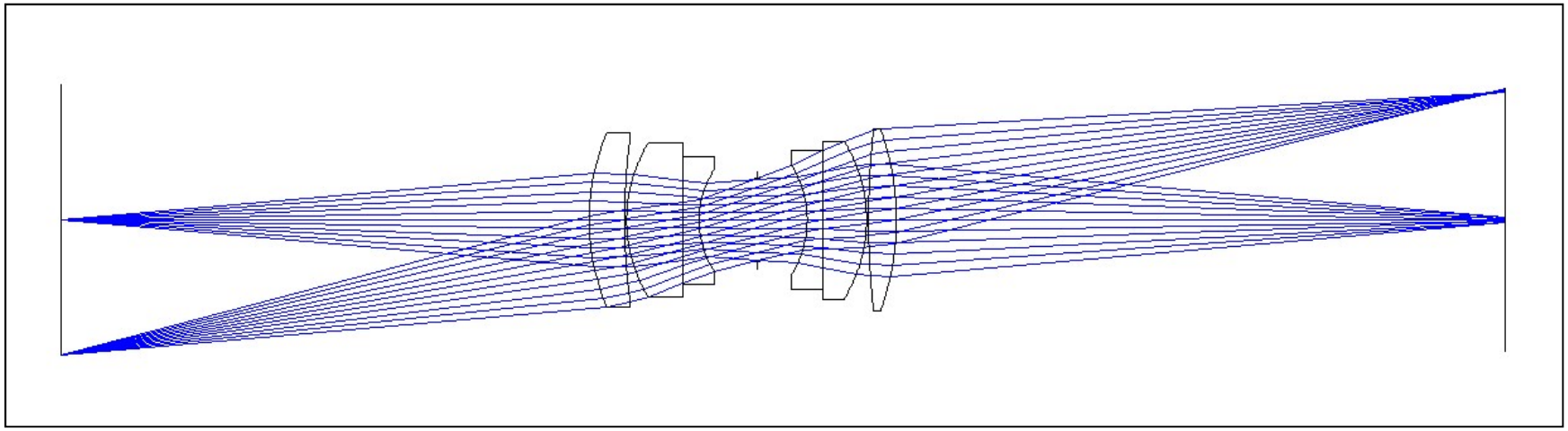
$$-k_0^2 \cdot E_{0x} \cdot \nabla S \cdot \nabla S + E_{0x} \cdot \nabla^2 S \cdot ik_0 + 2 \cdot \nabla E_{0x} \cdot ik_0 \cdot \nabla S + \nabla^2 E_{0x} = -k^2 \cdot E_{0x}$$

▪ Illustrating the intensity law ▪

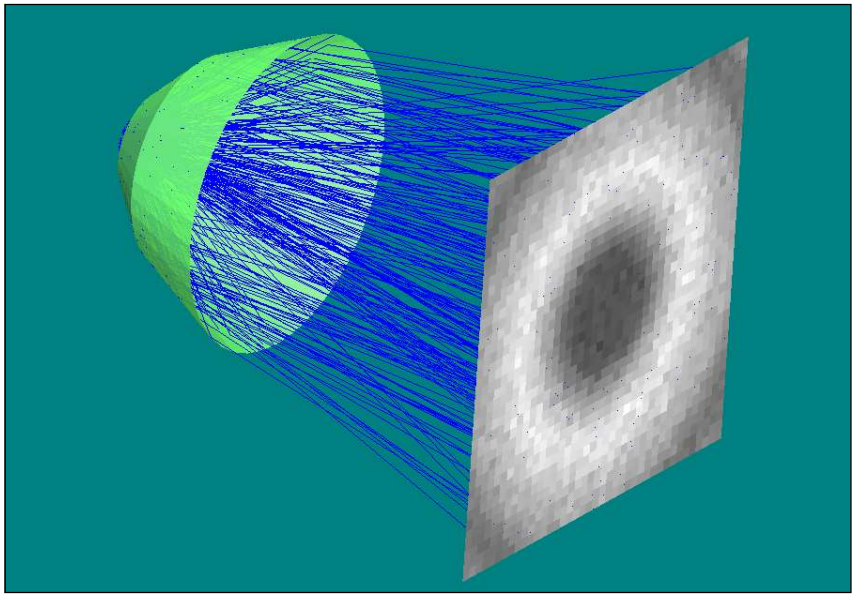




▪ Application of geometrical optics ▪

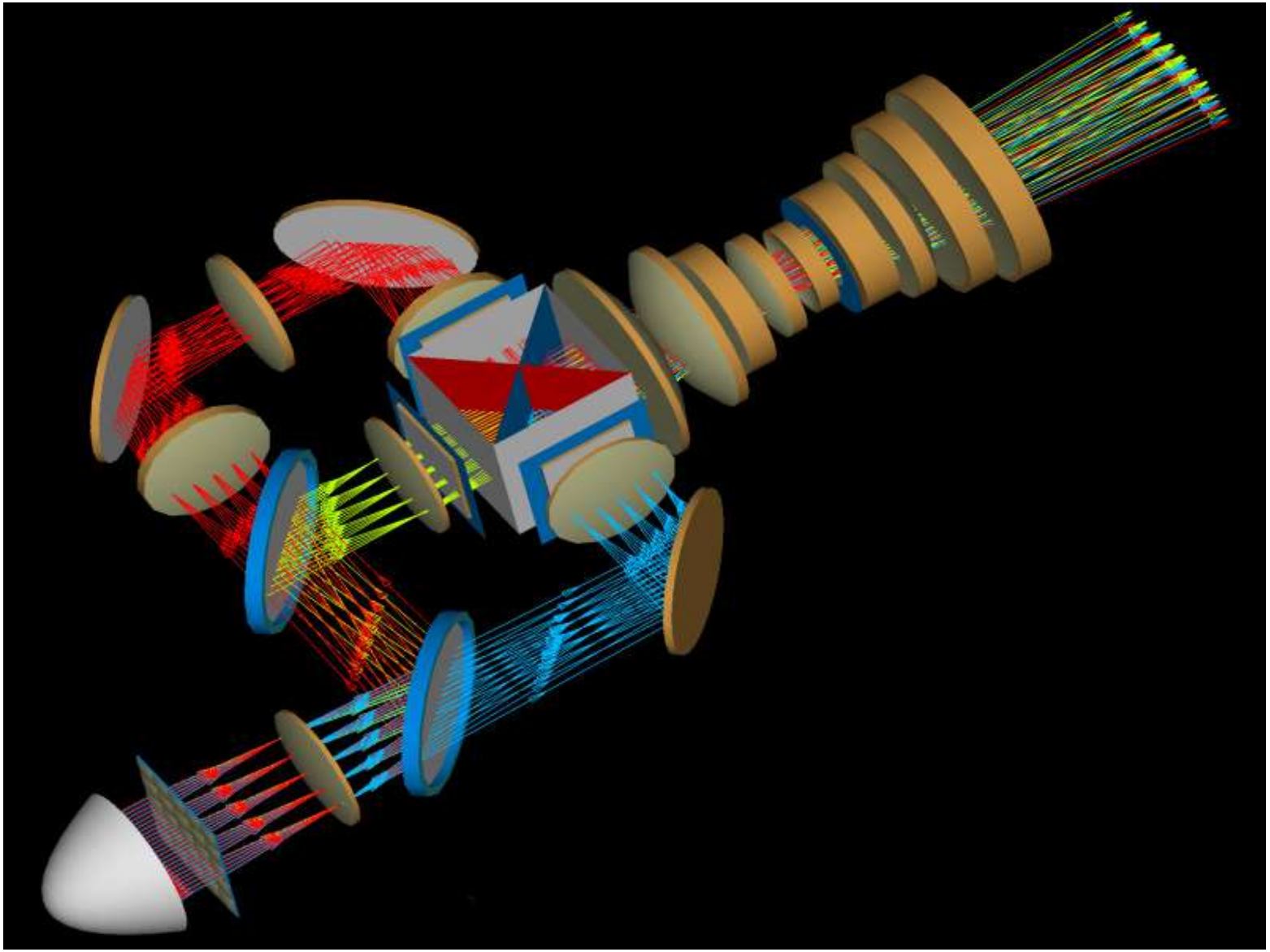


imaging systems

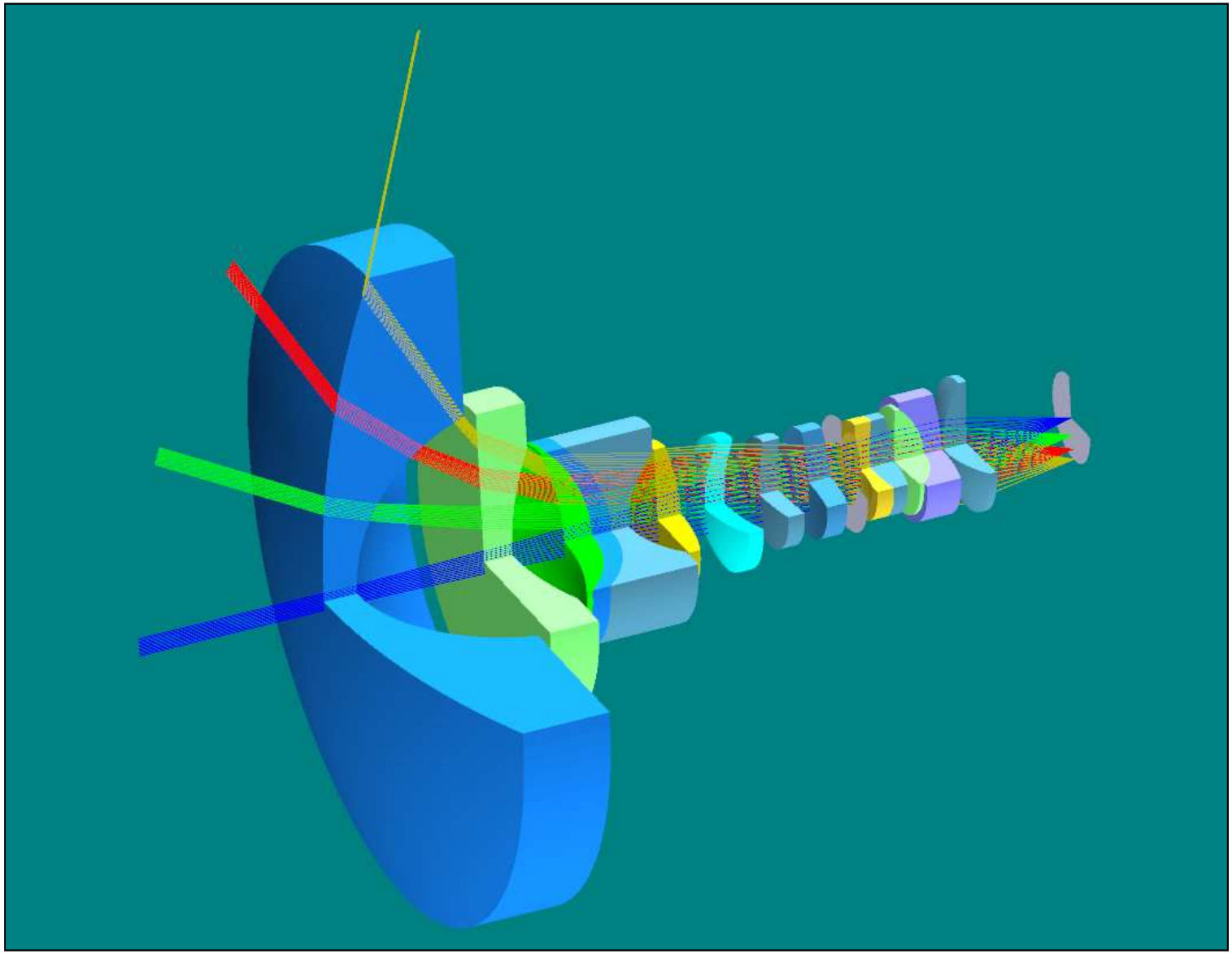


illumination systems

▪ Video projector ▪



▪ Wide-field objective (210°) ▪



▪ 193 nm lithography ▪



▪ 193 nm lithography ▪



Zeiss Starlith™ 900

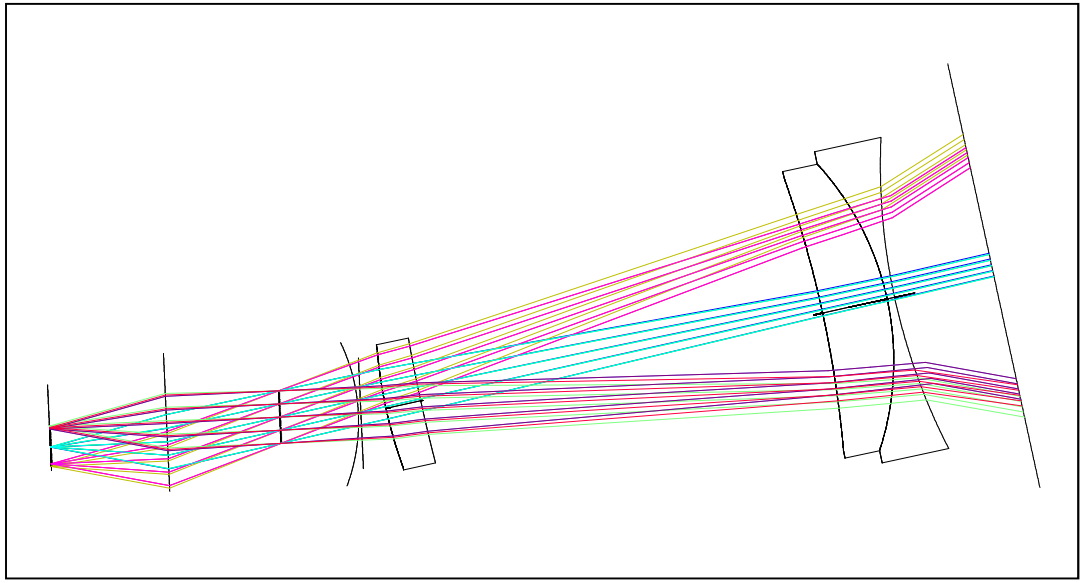
m = −1/4 ×

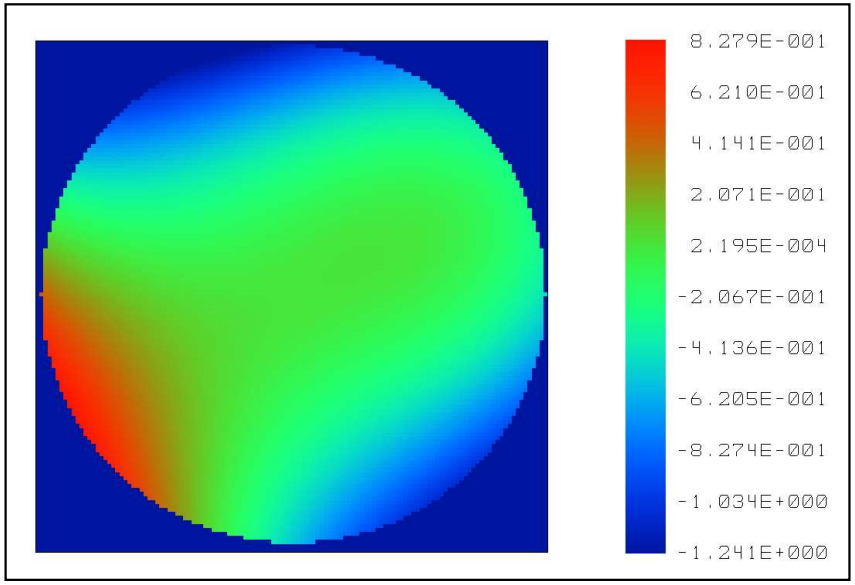
NA 0.63

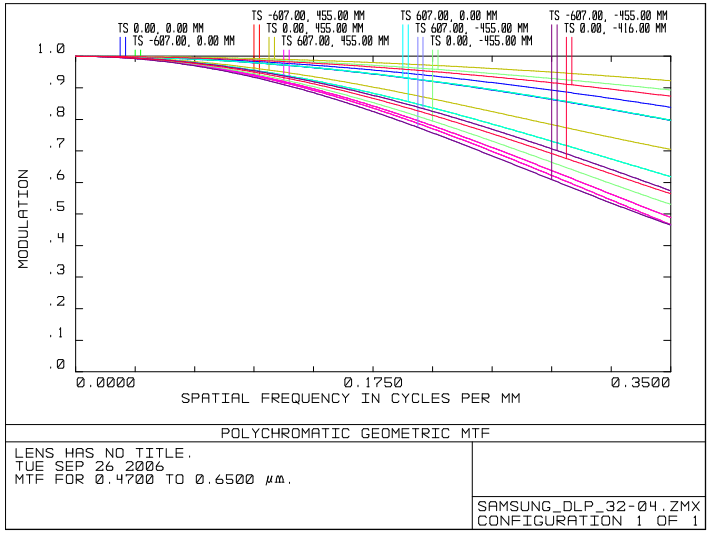
image field 22 × 22 mm

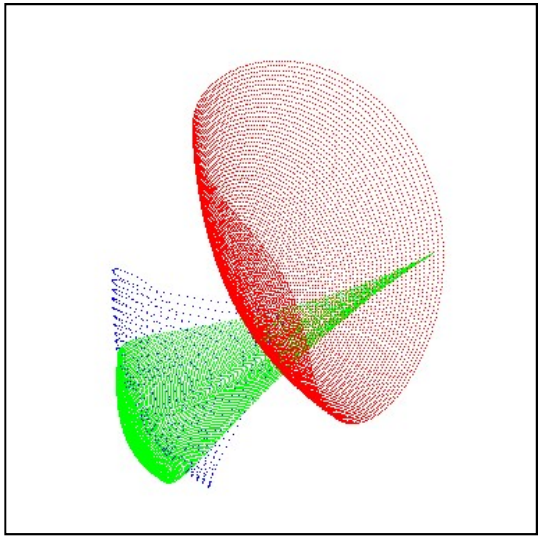
line width < 100 nm

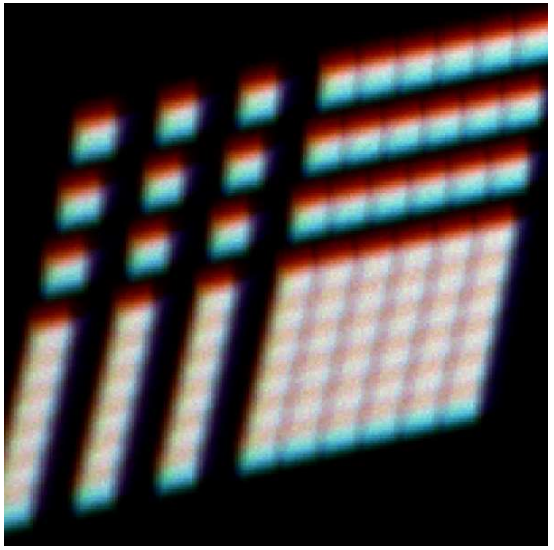
▪ Projector add-on lens ▪



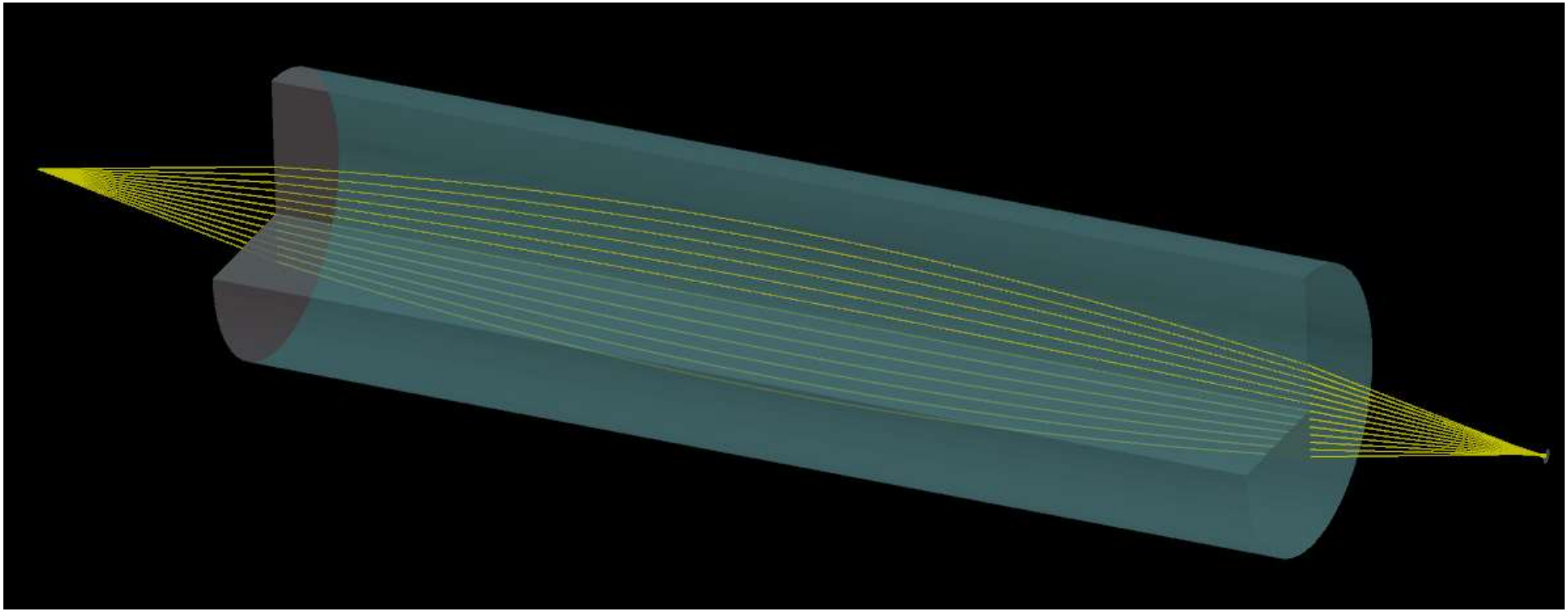






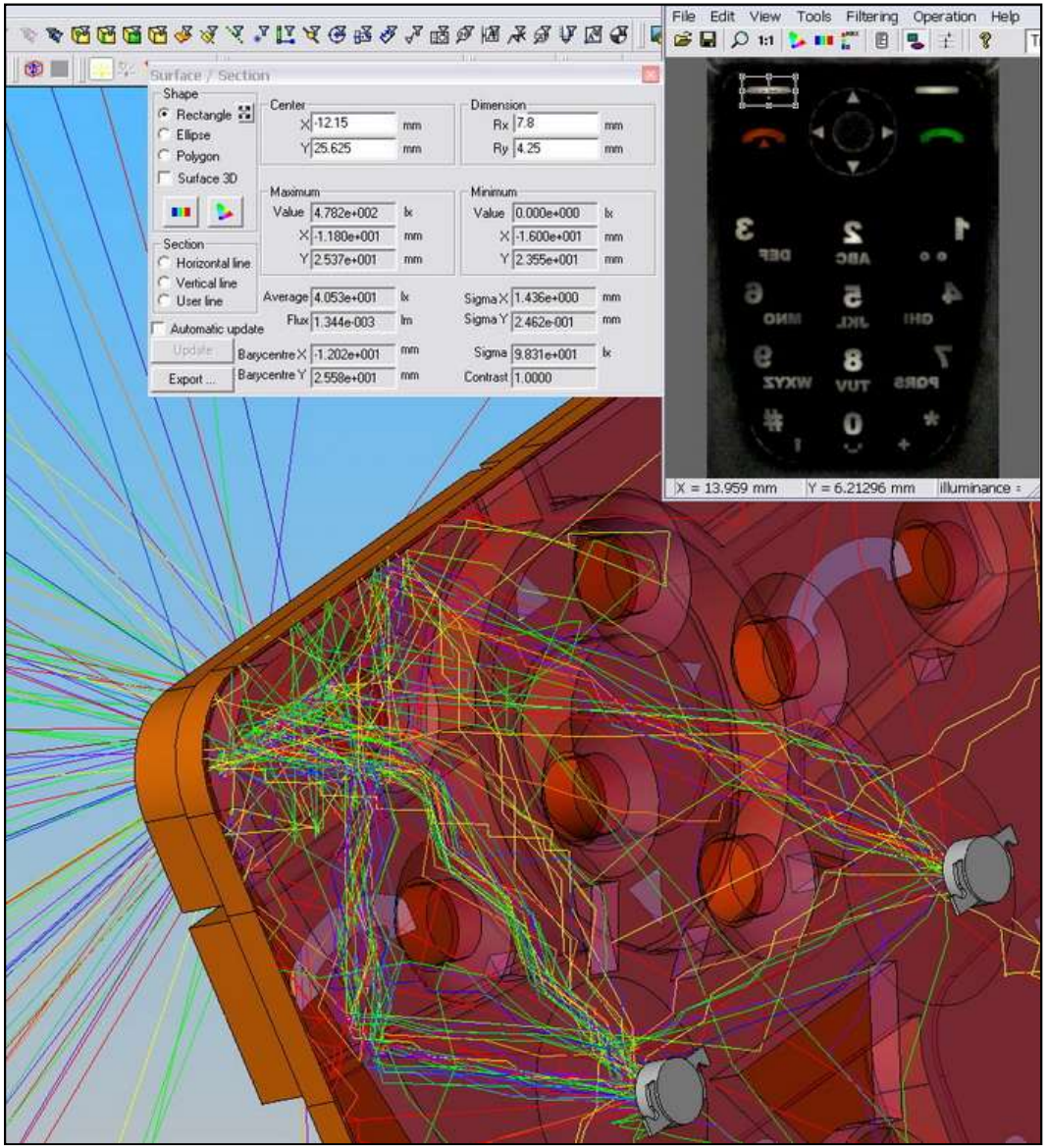


▪ Inhomogeneous media ▪

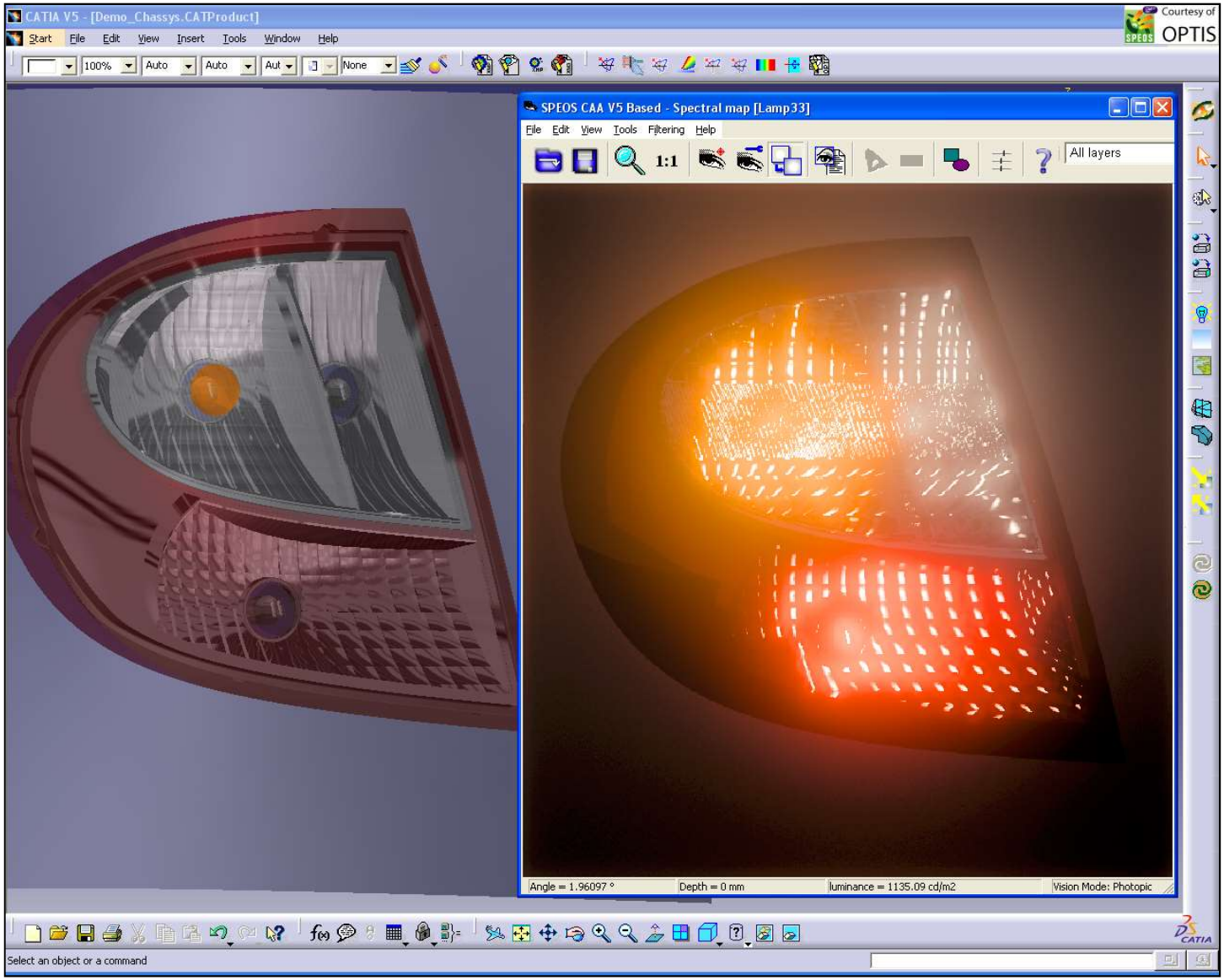


SELFOC lens

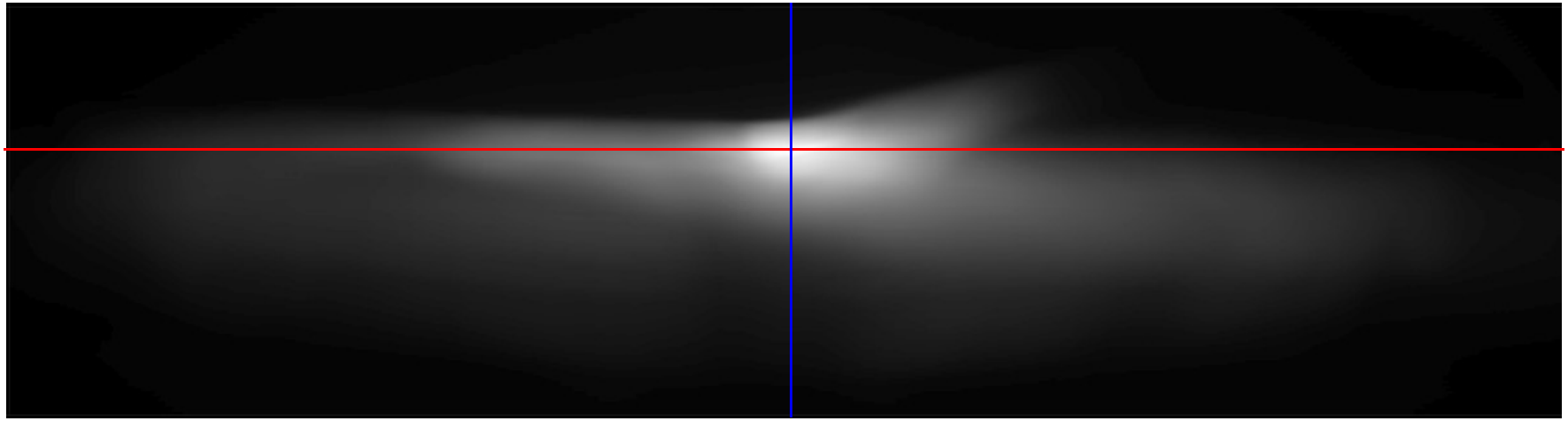
▪ Phone keypad illumination ▪

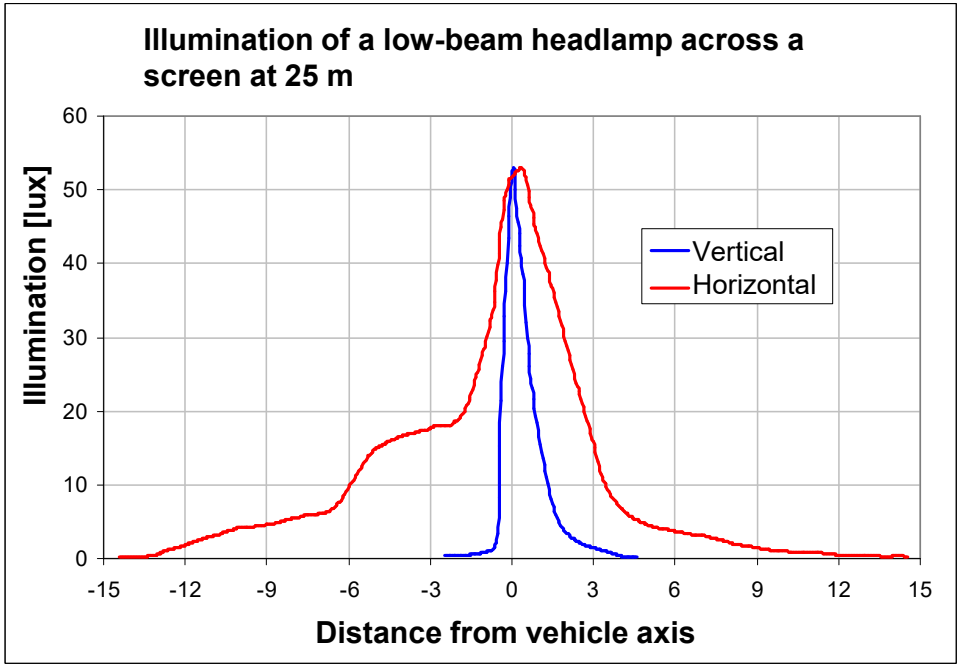


▪ Vehicle illumination technology ▪



▪ Low-beam headlamp ▪



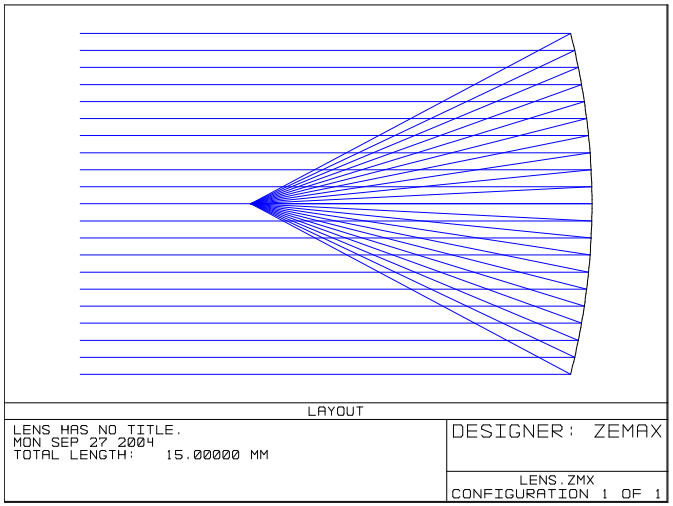


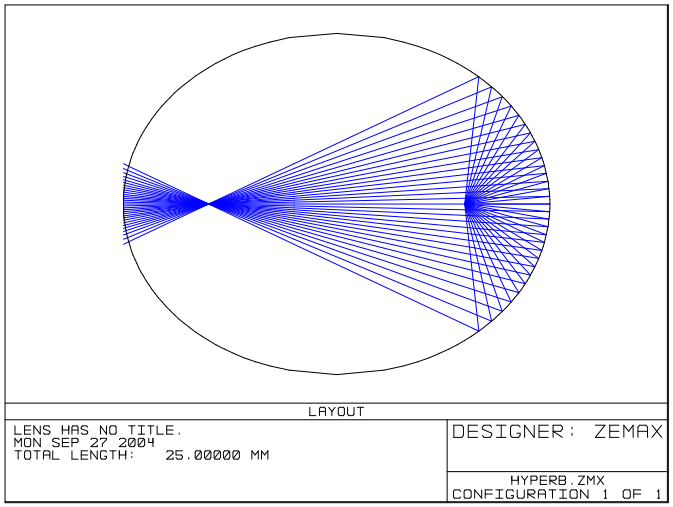
▪ Computer-generated imagery ▪



▪ Descartes surfaces ▪

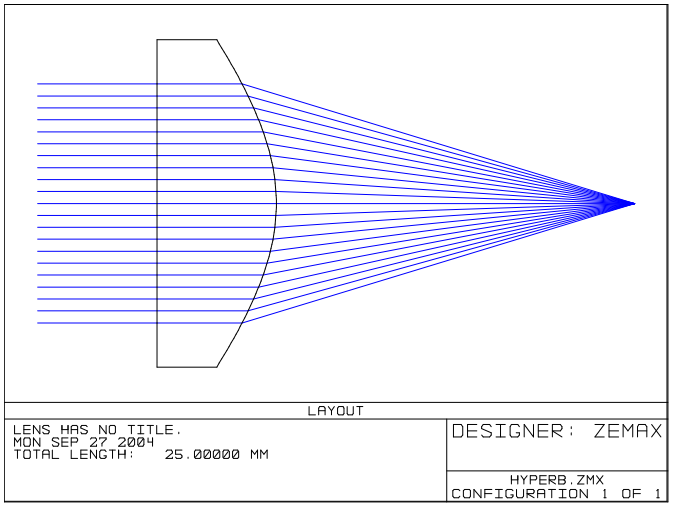
Paraboloid mirror

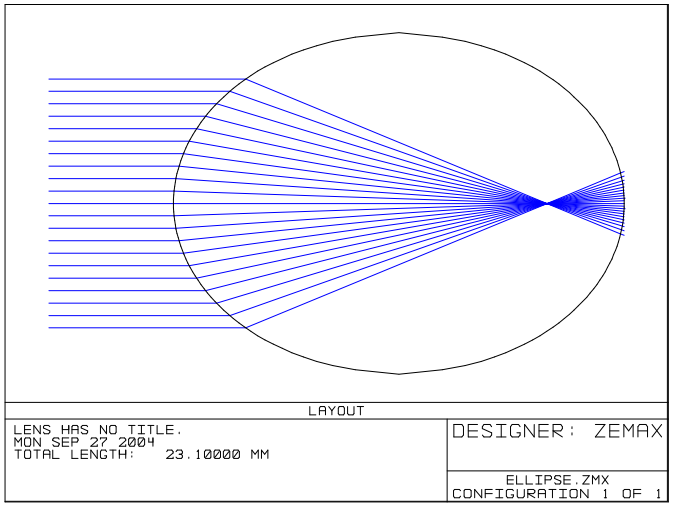




Ellipsoid mirror

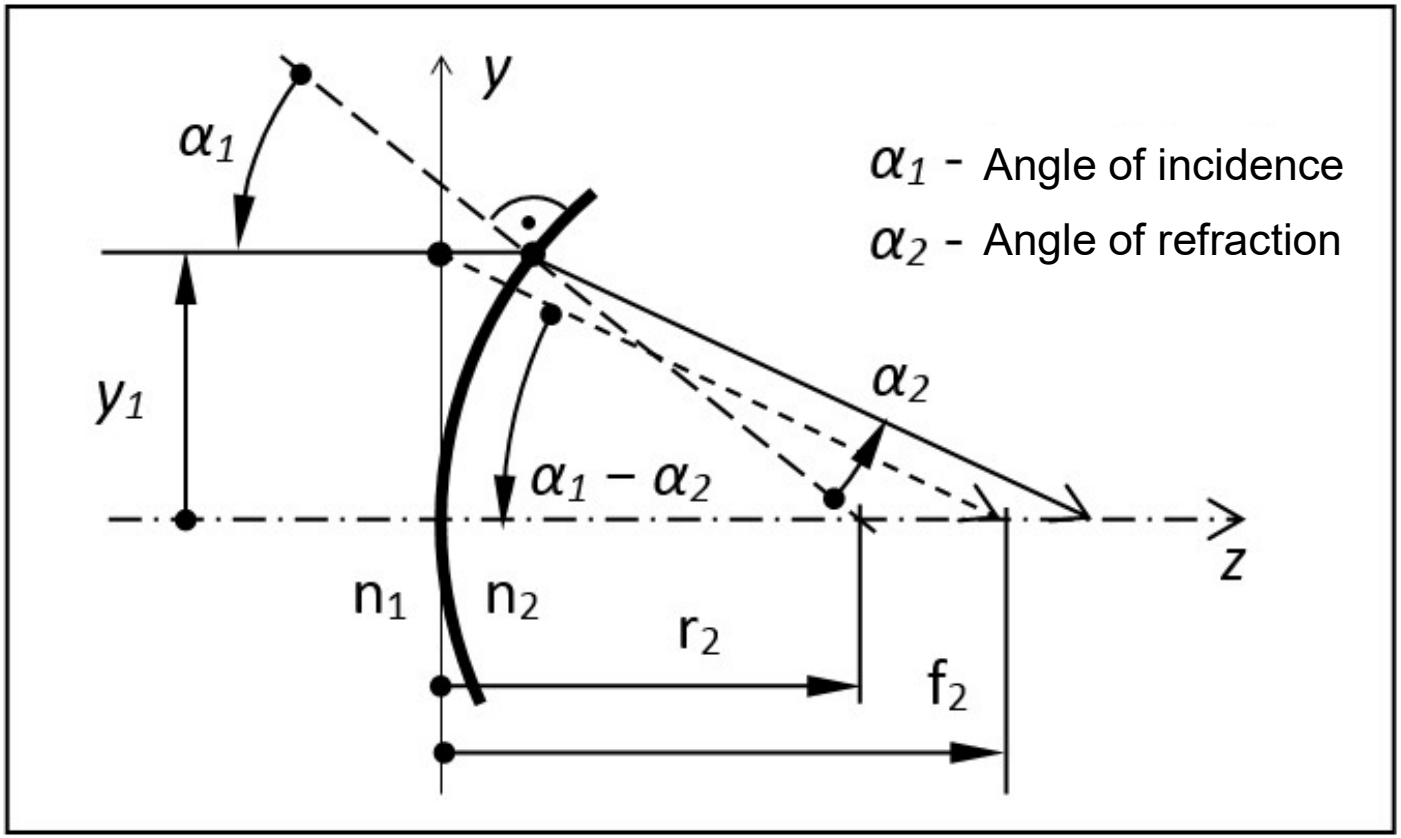
Hyperboloid lens



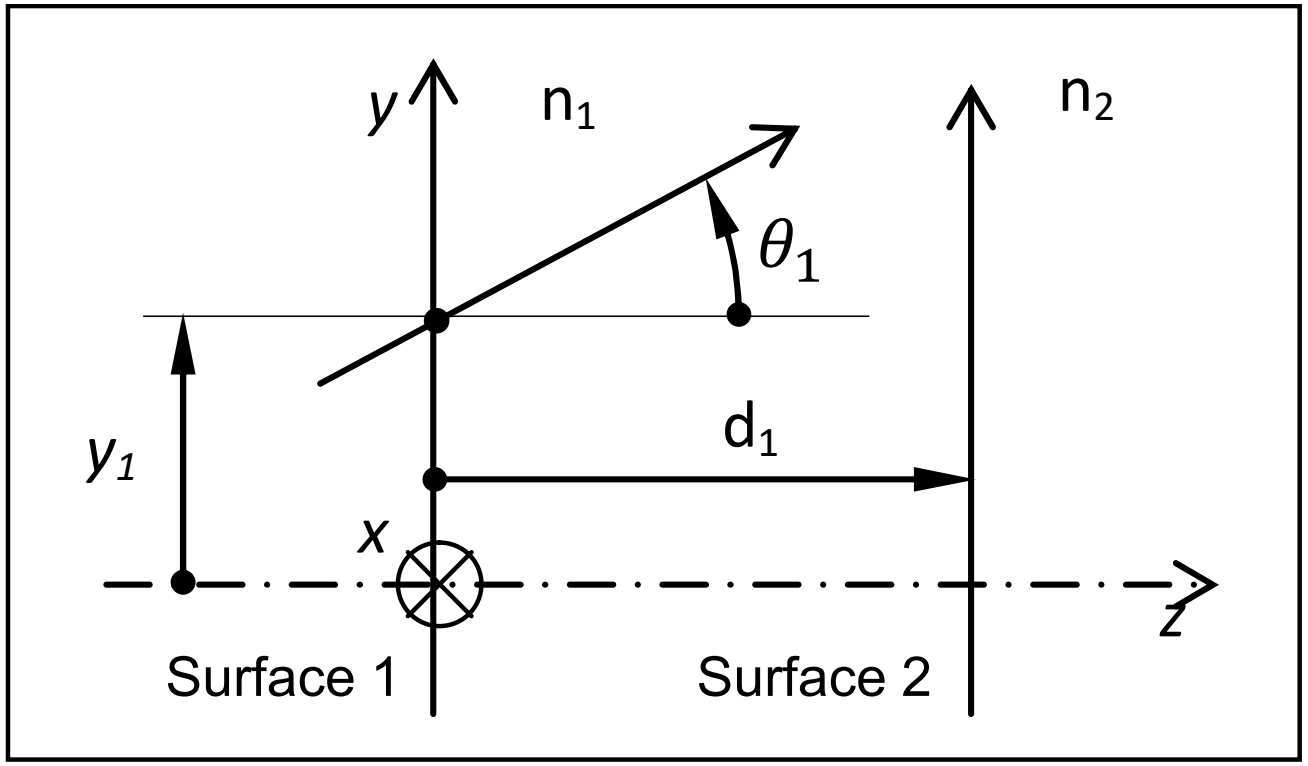


Ellipsiod lens

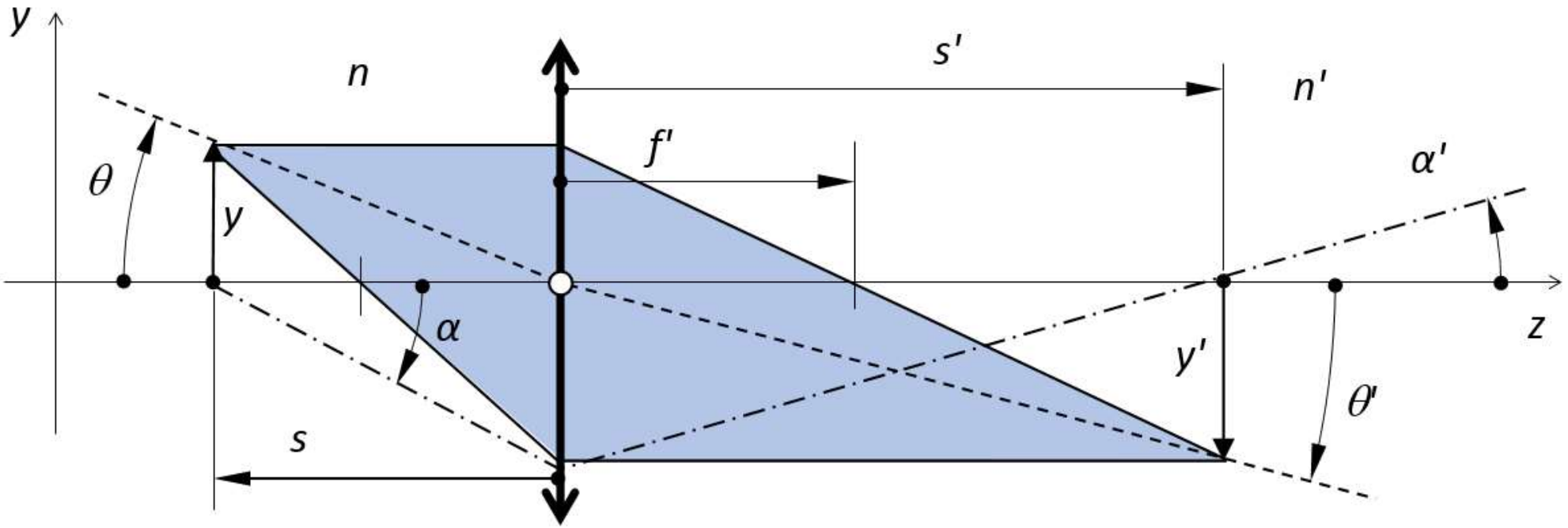
▪ Determining the focal length ▪



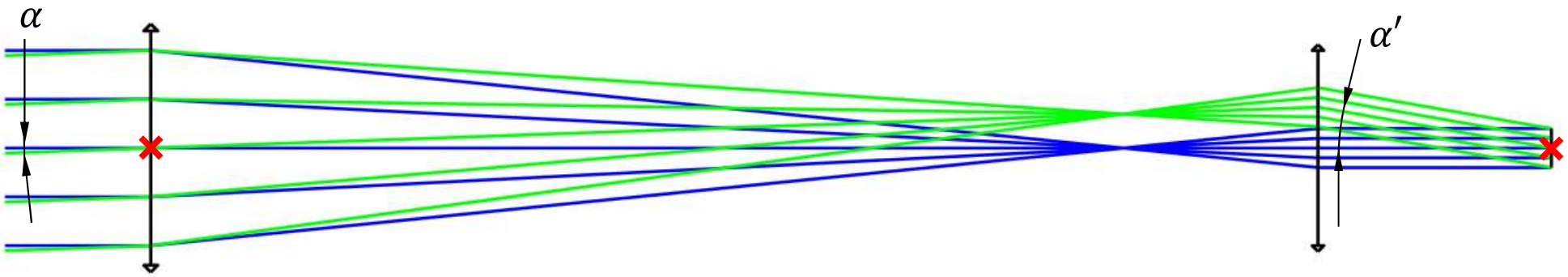
▪ Paraxial ray parameters ▪



▪ Characteristics of thin lenses ▪



▪ Angular magnification of telescopes ▪



▪ Determining a principal plane ▪

